

Non-parametric statistical hypothesis tests

Location tests

One sample or paired data

Sign test

Wilcoxon signed rank test

Two samples

Mann-Whitney U-test

Multiple samples

Kruskall-Wallis test (ANOVA1)

Friedman test (ANOVA"2")

Contingency table test for categorical data

Distribution tests

One sample Kolmogorov-Smirnov test

Two-sample Kolmogorov-Smirnov test

Goodness of fit test

Location test – one sample or paired data

Parametric version

$$X \sim N(\mu, \sigma^2)$$

$$H_0: \mu = \mu_0 \quad \text{vs.} \quad H_1: \mu \neq \mu_0, \mu > \mu_0, \mu < \mu_0$$

$$\text{Test statistic:} \quad T = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} \sim t_{n-1}$$

MATLAB: **ttest**

Non-parametric sign test: **signtest**

MATLAB-demo

one sample: test median for X , m_x

paired data: test median for $X - Y$, $m_x - m_y$

No assumptions on distribution

$$H_0: m = m_0 \Leftrightarrow P(X > m_0) = \frac{1}{2}$$

$$\text{Test statistic:} \quad S = \#\text{obs} > m_0 \sim \text{Bin}(n, \frac{1}{2})$$

Exact test: binomial test

$$p_value = 2 \min \{P[\text{Bin}(n, \frac{1}{2}) < S], P[\text{Bin}(n, \frac{1}{2}) > S]\} \quad \text{binocdf}$$

Approximate test: large sample normal approximation to binomial

$$S \sim \text{Bin}(n, \frac{1}{2}) \Rightarrow E[S] = \frac{n}{2}, V[S] = n \frac{1}{2} (1 - \frac{1}{2})$$

$$Z_0 = \frac{S - E[S]}{\sqrt{V[S]}} = \frac{S - \frac{n}{2}}{\sqrt{n/4}} \approx N(0,1)$$

$$p_value = 2 P[N(0,1) > |Z_0|] \quad \text{normcdf}$$

Simple test, but with low power = $P(\text{reject } H_0 | H_0 \text{ is false})$

Location test – one sample or paired data

Non-parametric Wilcoxon signed rank test: **signrank**

MATLAB-demo

Higher power than sign test but (somewhat) more complicated

No assumptions on distribution

$$H_0: m = m_0$$

Test statistic:

- For observations $x_1, \dots, x_n$ calculate $|x_i - m_0|$ and $\text{sign}(x_i - m_0)$
- Discard observations x_i if $x_i = m_0 \rightarrow n_r$ observations remaining
- Sort $|x_i - m_0|$ in increasing order
- Apply ranks $R_i = 1, 2, \dots, n_r$ to the sorted list
- $W = \sum_{i=1}^{n_r} \text{sign}(x_i - m_0) \cdot R_i$

W has known, but complicated distribution

$$E[W] = 0, \quad V[W] = \frac{n_r(n_r+1)(2n_r+1)}{6}$$

Approximate test: large sample normal approximation to W

$$Z_0 = \frac{W - E[W]}{\sqrt{V[W]}} = \frac{W}{\sqrt{\frac{n_r(n_r+1)(2n_r+1)}{6}}} \approx N(0,1)$$

$$p_value = 2 P[N(0,1) > |Z_0|] \quad \text{normcdf}$$

Alternative (equivalent) test statistic: (used by MATLAB **signrank**)

Decompose $W = W^+ + W^-$ (positive and negative rank sums)

$$T = \min(W^+, |W^-|)$$

$$E[T] = \frac{n_r(n_r+1)}{4}, \quad V[W] = \frac{n_r(n_r+1)(2n_r+1)}{24}$$

$$Z_0 = \frac{T - E[T]}{\sqrt{V[T]}} = \frac{T - \frac{n_r(n_r+1)}{4}}{\sqrt{\frac{n_r(n_r+1)(2n_r+1)}{24}}} \approx N(0,1)$$

$$p_value = 2 P[N(0,1) > |Z_0|] \quad \text{normcdf}$$

Location test – two samples

Parametric version

$$X \sim N(\mu_x, \sigma_x^2), \quad Y \sim N(\mu_y, \sigma_y^2)$$

$$H_0: \mu_x = \mu_y$$

Test statistic: $T = \frac{\bar{x} - \bar{y}}{\sqrt{\frac{s_x^2}{n_x} + \frac{s_y^2}{n_y}}} \sim t_{n_x + n_y - 2}$ MATLAB: **ttest2**

Non-parametric Mann-Whitney U test (Wilcoxon rank sum): **ranksum**

MATLAB-demo

No assumptions on distribution

$$H_0: m_x = m_y \Leftrightarrow P(X > Y) = P(Y > X)$$

Test statistic: $U = \sum_{i=1}^{n_x} \sum_{j=1}^{n_y} S(x_i, y_j)$ where

$$S(X, Y) = \begin{cases} 1 & X > Y \\ \frac{1}{2} & X = Y \\ 0 & X < Y \end{cases}$$

i.e., U counts for each x-observation how many y-observations are smaller and sums these counts over all x-observations (total number of x “victories”)

U has known, but complicated distribution

$$E[U] = \frac{n_x n_y}{2}, \quad V[U] = \frac{n_x n_y (n_x + n_y + 1)}{12}$$

Approximate test: large sample normal approximation to U

$$Z_0 = \frac{U - E[U]}{\sqrt{V[U]}} = \frac{U - \frac{n_x n_y}{2}}{\sqrt{\frac{n_x n_y (n_x + n_y + 1)}{12}}} \approx N(0, 1)$$

$$p_value = 2 P[N(0, 1) > |Z_0|] \quad \text{normcdf}$$

Alternative (equivalent) test statistic: (used by MATLAB **ranksum**)

$$W = \text{ranksum}(X) = \sum_{i=1}^{n_x} \text{rank}(x_i)$$

i.e., sum of all x-ranks when x- and y-observations are sorted together in increasing order and given ranks 1: (n_x + n_y)

$$W = U + \frac{n_x(n_x + 1)}{2}, \quad E[W] = E[U] + \frac{n_x(n_x + 1)}{2}, \quad V[W] = V[U]$$

Location test – multiple samples

Parametric version

ANOVA 1, MATLAB: **anova1**

$X_i \sim N(\mu_i, \sigma^2), \quad i = 1, \dots, g$

$H_0: \forall \mu_i = \mu$

Test statistic: $F = \frac{SSB/dfB}{SSW/dfW} = \frac{\sum_{i=1}^g \sum_{j=1}^{n_i} (\bar{X}_i - \bar{X})^2 / (g-1)}{\sum_{i=1}^g \sum_{j=1}^{n_i} (X_{ij} - \bar{X}_i)^2 / (N-g)} \sim F_{g-1, N-g}$

Non-parametric Kruskal-Wallis test : **kruskalwallis**

MATLAB-demo

No assumptions on distribution

Test for (column) factor effect

$H_0: \forall m_i = m, \quad i = 1, \dots, g$

$H_1: \exists i, j: m_i \neq m_j, \quad i, j = 1, \dots, g$

Data **rank**s are used instead of the actual numerical data

data $\{x_{ij}\} \rightarrow$ ranks $\{r_{ij}\} \in \{1, 2, \dots, N = \sum_{i=1}^g n_i\}$

i.e., all observations from all g populations are sorted together in increasing order and given its corresponding rank

Test statistic:

$$H = (N - 1) \frac{\sum_{i=1}^g n_i (\bar{r}_i - \bar{r})^2}{\sum_{i=1}^g \sum_{j=1}^{n_i} (r_{ij} - \bar{r})^2} \approx \chi_{g-1}^2 \quad \left(\frac{\text{"SSB"}}{\text{"SST"}} \right)$$

where

r_{ij} = rank (among all observations) of observation j from population i

$\bar{r}_i = \frac{\sum_{j=1}^{n_i} r_{ij}}{n_i}$ = average rank of all observations in population i

$\bar{r} = \frac{N+1}{2}$ = average rank of all observations

$p_value = P[\chi_{g-1}^2 > H]$ **chi2cdf**

H_0 rejected $\Rightarrow$ pairwise Mann-Whitney U tests to find out which populations differ in median value

Location test – multiple samples

Parametric version

ANOVA “2”, MATLAB: **anova2**

ANOVA “2” is often called “*Repeated measures ANOVA 1*” or “*Randomized block design ANOVA 1*” and is not a true two-factor ANOVA 2 because only the column factor is tested for.

The rows (often called blocks) correspond to each test object being subjected to each of the different “treatments” or “factor levels”.

It is not always possible to do an ANOVA 1 in this way because it requires that each test object can be “split into parts”, but when possible it gives a test for column factors with higher power because the nuisance effect of different test object (block) properties are removed.

$$X_{ij} \sim N(\mu_{ij}, \sigma^2), \quad i = 1, \dots, g \quad j = 1, \dots, n$$

$$\mu_{ij} = \mu + \tau_i + \beta_j$$

$$H_0: \forall \tau_i = 0$$

Test statistic:

$$F = \frac{SSB/dfB}{SSW/dfW} = \frac{\sum_{i=1}^g n (\bar{X}_{i\bullet} - \bar{X})^2 / (g-1)}{\sum_{i=1}^g \sum_{j=1}^n (X_{ij} - \bar{X}_{i\bullet} - \bar{X}_{\bullet j} + \bar{X})^2 / (n-1)(g-1)} \sim F_{g-1, (n-1)(g-1)}$$

Non-parametric Friedman test : **friedman**

MATLAB-demo

No assumptions on distribution

Test only for column factor effect (as explained above)

$$H_0: \forall m_i = m, \quad i = 1, \dots, g$$

For each test object (block, row) the data values for the g populations are sorted in increasing order and given corresponding ranks from 1 to g

Thus: data $\{x_{ij}\} \rightarrow$ ranks $\{r_{ij}\} \in \{1, 2, \dots, g\}$

Test statistic: $Q = \frac{12n}{g(g+1)} \sum_{i=1}^g (\bar{r}_{i\bullet} - \bar{r})^2 \approx \chi_{g-1}^2$

where

$$\bar{r}_{i\bullet} = \frac{\sum_{j=1}^n r_{ij}}{n} = \text{average rank of all observations in population } i$$

$$\bar{r} = \frac{g+1}{2} = \text{average rank for all } r_{ij}$$

$$p_value = P[\chi_{g-1}^2 > Q] \quad \text{chi2cdf}$$

Categorical data

Non-parametric contingency table test : **crosstab**

MATLAB-demo

No assumptions on distribution

Categorical data are **counts** of how many of the observations (test objects) belong to specific categories (factor levels) of factors A and B.

This looks at first sight somewhat like a two-factor ANOVA 2, but here the entries in the table are category membership counts and not actual numerical data values.

The counts can be represented in a **contingency table**:

		← factor B levels →			
		B ₁	B ₂	...	B _c
↑ factor A levels ↓	A ₁				
	A ₂				
	...				
	A _r				

Observed counts: $O(i, j) = \# \text{ obs for category combination } (A_i, B_j)$

H_0 : factors A and B are independent

Expected counts (given H_0):
$$E(i, j) = \frac{\sum_{j=1}^c O(i, j) \cdot \sum_{i=1}^r O(i, j)}{\sum_{i=1}^r \sum_{j=1}^c O(i, j)}$$

i.e., (rowsum for factor A level A_i counts) times
(column sum for factor B level B_j counts) divided by
total number of observations

Test statistic:
$$T = \sum_{i=1}^r \sum_{j=1}^c \frac{[O(i, j) - E(i, j)]^2}{E(i, j)} \approx \chi_{(r-1)(c-1)}^2$$

$p_value = P[\chi_{(r-1)(c-1)}^2 > T]$ **chi2cdf**

Parametric analogy

As mentioned above the contingency table somewhat resembles the data set for ANOVA 2. The contingency table test for independent factors then correspond to the ANOVA 2 test for no systematic interaction between factors and the degrees of freedom, $(\#rows - 1)(\#columns - 1)$, is also the same. But, again, the contingency table entries are counts of category combination memberships and not numerical “real data” as in ANOVA 2.

Distribution test

One sample Kolmogorov-Smirnov test: **kstest**

MATLAB-demo

The usual quantile-quantile QQ-plot (MATLAB **qqplot**) is as known a visualization of quantiles for a data set against quantiles for a supposed (hypothesized) theoretical distribution. Visual inspection gives some insight as to whether it is plausible that the data sample comes from a random variable with the supposed distribution (if the quantile pairs are approximately lying on a straight line).

The Kolmogorov-Smirnov test quantifies the visual inspection into a formal hypothesis test. A relatively large sample is required for the test to be reliable.

H_0 : The sample data $x_i, i = 1, \dots, n$ comes from random variable X with cdf (cumulative density function) $F(x)$ with **known** parameters

The data set observations are sorted in increasing order:

$$x_1 \leq x_2 \leq \dots \leq x_n$$

Then the empirical data cdf is:

$$\hat{F}(x) = \frac{\#\text{obs} \leq x}{n} = \frac{1}{n} \sum_{i=1}^n 1_{x_i \leq x} \quad \text{where} \quad 1_{x_i \leq x} = \begin{cases} 1 & x_i \leq x \\ 0 & x_i > x \end{cases} \quad \text{cdfplot}$$

Test statistic: $D = \max_x |\hat{F}(x) - F(x)|$
 i.e., maximal vertical difference between empirical and theoretical cdf's

Critical value for level of significance α : $c(\alpha) = \sqrt{\frac{-\frac{1}{2} \log \frac{\alpha}{2}}{n}}$
 $\alpha = 0.05 \Rightarrow c(\alpha) \cong 1.36/\sqrt{n}$

Test: $D > c(\alpha) \Rightarrow \text{reject } H_0$

Distribution test

Two sample Kolmogorov-Smirnov test: **kstest2**

MATLAB-demo

The two sample problem is to decide whether it is plausible that two data sets are drawn from random variables with identical distributions.

The QQ-plot can also in this case give some insight by plotting the empirical quantiles for the two data sets against each other and making a visual inspection as to whether the quantile pairs are approximately lying on a straight line

The two sample Kolmogorov-Smirnov test quantifies the visual inspection into a formal hypothesis test.

H_0 : The random variable (population) X_1 giving sample data x_{1i} , $i = 1, \dots, n$ and the random variable X_2 giving sample data x_{2i} , $i = 1, \dots, m$ have the same distribution (and therefore the same cdf)

The data set observations are sorted in increasing order:

$$x_{11} \leq x_{12} \leq \dots \leq x_{1n} \quad \text{and} \quad x_{21} \leq x_{22} \leq \dots \leq x_{2m}$$

Empirical data cdf's:

$$\hat{F}_1(x) = \frac{\#\text{obs} \leq x}{n} = \frac{1}{n} \sum_{i=1}^n 1_{x_{1i} \leq x} \quad \text{where} \quad 1_{x_{1i} \leq x} = \begin{cases} 1 & x_{1i} \leq x \\ 0 & x_{1i} > x \end{cases} \quad \text{cdfplot}$$

$$\hat{F}_2(x) = \frac{\#\text{obs} \leq x}{m} = \frac{1}{m} \sum_{i=1}^m 1_{x_{2i} \leq x} \quad \text{where} \quad 1_{x_{2i} \leq x} = \begin{cases} 1 & x_{2i} \leq x \\ 0 & x_{2i} > x \end{cases} \quad \text{cdfplot}$$

Test statistic: $D = \max_x |\hat{F}_1(x) - \hat{F}_2(x)|$
 i.e., maximal vertical difference between empirical cdf's

Critical value for level of significance α : $c(\alpha) = \sqrt{-\frac{1}{2} \log \frac{\alpha}{2} \left(\frac{1}{n} + \frac{1}{m} \right)}$

$$\alpha = 0.05 \Rightarrow c(\alpha) \cong 1.36 \sqrt{\left(\frac{1}{n} + \frac{1}{m} \right)}$$

Test: $D > c(\alpha) \Rightarrow \text{reject } H_0$

Goodness Of Fit test (Pearson's Chi-Squared test), chi2gof

MATLAB-demo

The Goodness Of Fit test is a **distribution test** like the Kolmogorov-Smirnov tests, but in this (in practice more often occurring) case the parameters of the hypothesized distribution are unknown and must thus be estimated from the data.

This means that the GOF test examines the plausibility as to whether the data come from a specific type of distribution, e.g. from an exponential or log-normal distribution.

H_0 : The sample data x_i , $i = 1, \dots, n$ are coming from a random variable, X , with a specified distribution type with probability density function $f(x)$ with unknown parameters

Estimation of unknown parameters

The unknown parameters are estimated from the data sample using, e.g., Maximum Likelihood estimation. The number of parameters, N_{par} , to be estimated depends on the hypothesized distribution type. For example an exponential distribution has just one parameter: $\lambda = 1/\mu$, while a log-normal distribution has two: (μ, σ^2)

The estimation of unknown parameters yields a by now fully specified, hypothesized pdf: $\hat{f}_{H_0}(x)$

Binning

Now the domain of $\hat{f}_{H_0}(x)$ (e.g., for exponential distribution $x \in [0, \infty[$), is split up into a number of bins, N_{bins} . For each bin $i = 1, \dots, N_{\text{bins}}$ the expected number of observations (given H_0) is calculated as

$$E_i = \int_{\text{bin}_i} \hat{f}_{H_0}(x) dx$$

The GOF test is most reliable if bins are chosen such that

$$\forall i, j: E_i \cong E_j \quad \text{and} \quad \forall i: E_i > 5$$

Test statistic

The test statistic compares the expected number of observations, E_i , in each bin with the observed number of observations, O_i :

$$T = \sum_{i=1}^{N_{\text{bins}}} \frac{(O_i - E_i)^2}{E_i} \approx \chi_{df}^2 \quad \text{where} \quad df = N_{\text{bins}} - N_{\text{par}} - 1$$
$$p_value = P[\chi_{df}^2 > T] \quad \text{chi2cdf}$$